UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

1594589560

PHYSICS **5054/02**

Paper 2 Theory **October/November 2008**

1 hour 45 minutes

Candidates answer on the Question Paper.

Additional Materials: Answer Booklet/Paper.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.

Section B
Answer any **two** questions.
Write your answers on the lined pages provided and, if necessary, continue on the separate answer paper provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
Q9	
Q10	
Q11	
Total	

This document consists of **16** printed pages.

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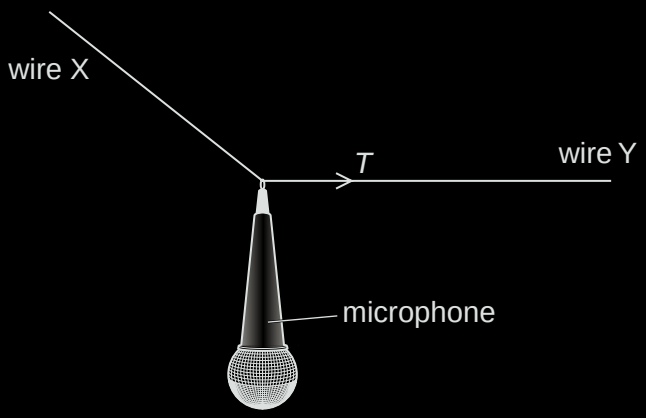


Fig. 1.1

(a) The tension *T* in wire Y is 8.0 N. Use a vector diagram to determine the magnitude and the direction of the resultant of *W* and *T.*

magnitude = ...

direction = ...
[4]

(b) The microphone is at rest. State the tension in wire X.

tension in X = ... [1]

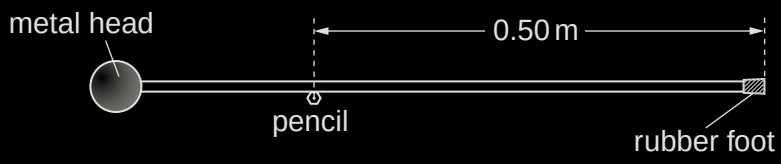


Fig. 2.1

(a) State the distance between the centre of mass of the walking-stick and the end of the rubber foot.

distance = ... [1]

(b) The pencil is moved along the walking-stick towards its rubber foot. State and explain the motion of the walking-stick.

.....

.....

.....

..... [3]

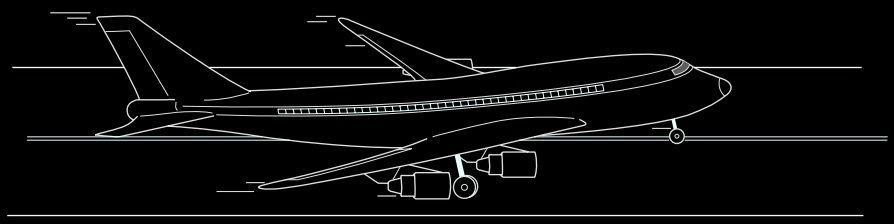


Fig. 4.1

(i) On Fig. 4.1, draw arrows to show the directions and the relative magnitudes of these forces. [1]

(ii) State what causes the smaller of these two forces.

..... [1]

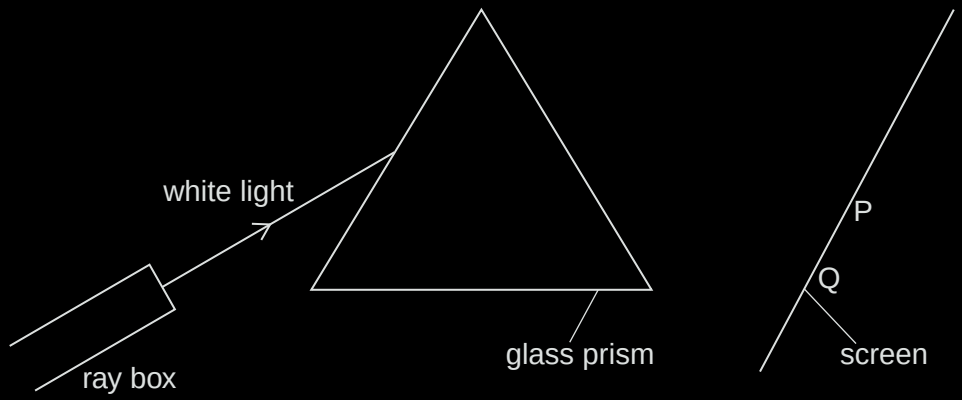


Fig. 6.1

(a) State the colour of the light at end P of the spectrum.

..... [1]

(b) State whether the value of each of these properties for blue light is greater than, equal to or less than the value for red light.

(i) speed in a vacuum ..... [1]

(ii) wavelength ..... [1]

(c) Fig. 6.2 shows the ray passing through a red filter before it reaches the prism.

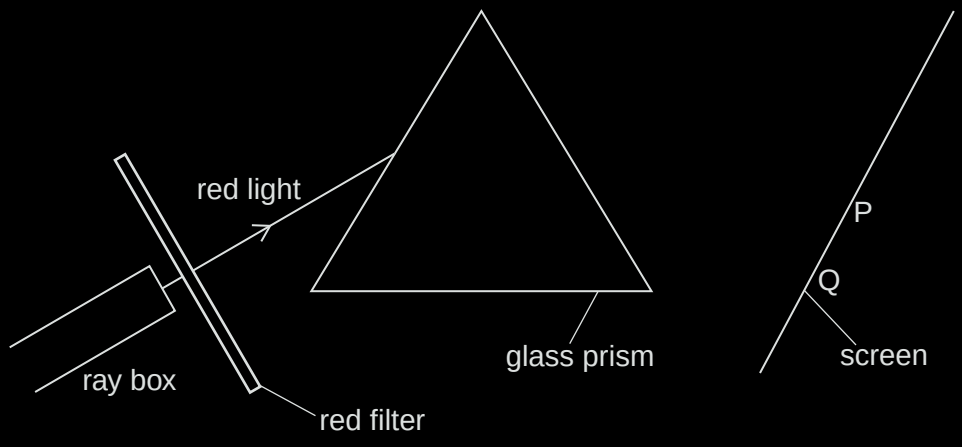


Fig. 6.2

Complete Fig. 6.2 to show the ray of red light passing through and emerging from the prism. [2]

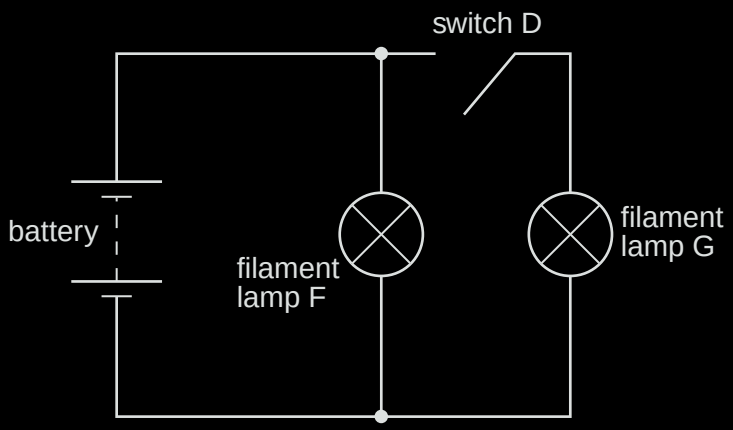


Fig. 7.1

When switch D is open, the battery supplies a current of 4.6 A.

Complete the table of Fig. 7.2.

	current supplied by battery / A	current in filament lamp F / A	current in filament lamp G / A
switch D open	4.6		
switch D closed			

Fig. 7.2 [3]

(c) Calculate the energy supplied by the battery as an electric charge of 200 C moves through the circuit.

energy supplied = ... [2]



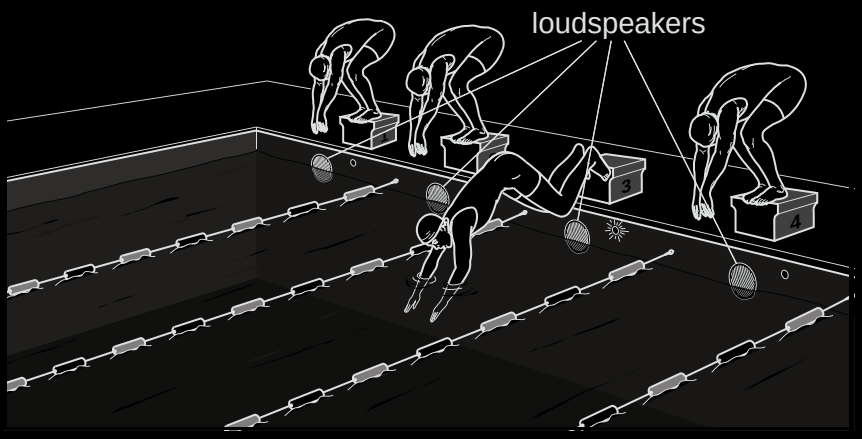


Fig. 9.1

The swimmer is called back by a loud, low-pitched sound from a loudspeaker positioned just at water level. The speed of sound in air is 330 m / s.

(a) **(i)** Describe how the loudspeaker causes sound to travel through the air. [3]

(ii) Explain, in terms of wave properties, what is meant by *loud* and *low-pitched*. [3]

(iii) The swimmer is 0.57 m from the loudspeaker when he hears the sound. Calculate the time taken for the sound to reach him through the air. [2]

(iv) Explain how the time taken differs when sound travels the same distance through air and through water. [2]

(b) The loudspeaker produces sound of frequency 0.20 kHz.

(i) Calculate the wavelength of this sound. [3]

(ii) Draw a diagram to show what is meant by the term *wavelength* when applied to a **longitudinal** wave such as sound. [2]

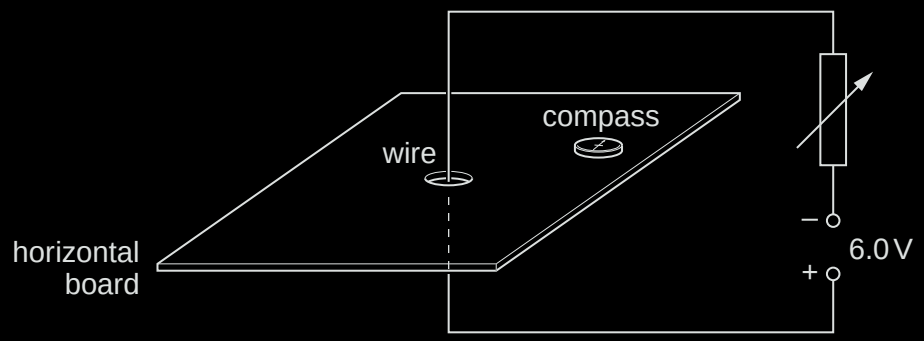


Fig. 10.1

The wire carries a current vertically upwards. A student moves a small compass around the board and plots the magnetic field lines due to the current.

(a) **(i)** Draw a diagram of the board as seen from above and mark on it the magnetic field lines due to the current. [3]

(ii) The current is increased. Describe how the magnetic field changes. [1]

(b) A 6.0 V power supply produces a current of 8.0 A in the wire. Calculate

(i) the total resistance of the circuit, [2]

(ii) the charge that flows through the wire in 2.0 minutes. [2]

(c) The north pole of a bar magnet is held on the left of the wire and the south pole of another bar magnet is held on the right, as shown in Fig. 10.2.

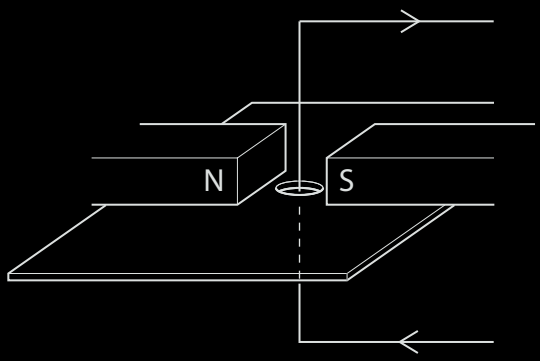


Fig. 10.2

(i) State the direction of the magnetic field at the wire due to these two poles. [1]

(ii) The wire is flexible. Describe the effect of this magnetic field on the current-carrying wire. [2]

(iii) The current in the wire is now reversed. State the effect of this on the wire. [1]

(iv) Describe how this effect is put to use in a d.c. motor. [3]

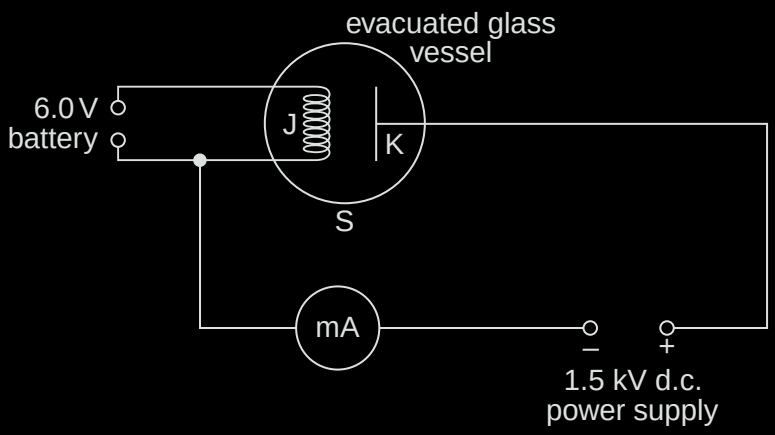


Fig. 11.1

(a) A 6.0 V battery is connected to J and the filament becomes white hot. The current from the battery is 1.6 A. Calculate the power supplied by the battery. [2]

(b) A milliammeter and a 1.5 kV d.c. power supply are connected in series between K and J. The positive terminal of the power supply is connected to K.

(i) The milliammeter registers a small current. Explain the presence of a current in this circuit despite the gap between J and K. [3]

(ii) State why the glass vessel must be evacuated. [1]

(iii) One pole of a bar magnet is brought close to the side S of the glass vessel and the current registered by the milliammeter decreases. Explain why this happens. [2]

(iv) The terminals of the 1.5 kV d.c. power supply are reversed. Explain how this affects the current in the milliammeter. [2]

Question 11 is continued on page 13.

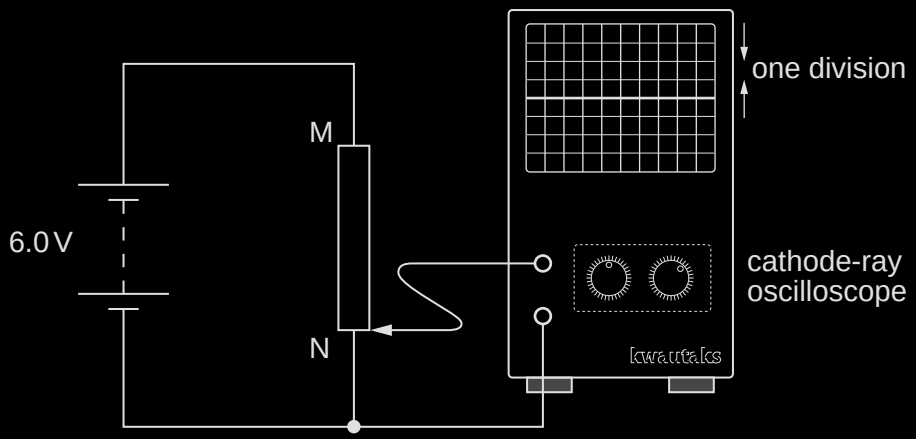


Fig. 11.2

The sliding contact is at N and the trace on the oscilloscope is a horizontal line passing through the centre of the screen.

(i) The timebase setting is 1.0 ms / div. Explain why the trace is a horizontal line. [1]

(ii) The Y-gain setting is 2.0 V / div. The sliding contact is moved at a slow, uniform rate from N to M. Describe in detail what happens to the trace on the screen. [3]

(iii) The Y-gain setting is now changed to 1.0 V / div and the trace disappears from the screen. State why this happens. [1]

